Observational Probability from Delay Queries: A Bridge from Sensor Streams to Predictive Operators

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Abstract

This paper bridges the foundational observational program of the preceding works with the computational theory of delay-coordinate methods for dynamical systems reconstruction. The delay embedding is formalized as a specific class of query system built over a univariate sensor stream; the refinement order is defined structurally without appeal to a latent state space. The delay query system is shown to satisfy all hypotheses of the observational extension theorem, providing it with a canonical probability space representation.

The minimal predictive query of Paper 2 characterizes when a delay embedding is sufficient for prediction: a delay embedding of dimension d and lag τ is predictively sufficient if and only if $d \geq m(\tau, P)$, the *predictive Markov order*. Takens' embedding theorem emerges as the deterministic special case of this probabilistic sufficiency criterion.

The geometric perspective — f-divergences, Fisher–Rao geometry, and the Rose condition — arises as the natural metric structure on the space of predictive laws. The predictive operator of Paper 3 specializes to a concrete Markov operator on $L^2(O_{Q^*}, \nu_{Q^*})$, whose local linear approximation by data-driven maps connects the theoretical framework to algorithms such as Dynamic Mode Decomposition and its probabilistic extensions.

1 Introduction and orientation

The observational program of the preceding papers develops the chain

$$\text{observable queries} \;\to\; \text{probability} \;\to\; \text{predictive state} \;\to\; \text{operator dynamics}.$$

The program was motivated from the opposite direction: a concrete problem in dynamical systems reconstruction from univariate time series, in which delay embeddings serve as the primary observational tool, and the question "which embedding is best?" ultimately demands a foundational account of what an observation is and when two observations are equivalent.

This note begins the return journey: given the foundational framework, what does a delay embedding look like within it, and what does the framework say about when one embedding is better than another?

Starting point. We must start somewhere. The primitive is a *sensor alphabet* $(X, \mathcal{X})$: a measurable space whose σ-algebra $\mathcal{X}$ encodes the observer's *reportable distinctions* — the yes/no questions about a single reading that the observer is capable of forming. This is not a claim about an external world that exists independently of observation; it is a minimal encoding of discriminative capacity, the minimum structure required to get off the ground.

Remark 1 (On the measurability assumption)**.** *The assumption that $(X, \mathcal{X})$ is a measurable space encodes finite Boolean closure of reportable distinctions together with closure under countable combinations. The finite part is relatively uncontroversial: if one can report A and B separately, one can report their union. The countable closure is a stronger idealization whose derivation from more primitive operational notions — roughly, that reportable distinctions are those approximable from finite observations — is an open problem deferred to future work. See Stopping Point 1 of the foundational program.*

The *observational horizon* is $\Omega = X^{\mathbb{Z}}$ with the product σ-algebra $\mathcal{F} = \mathcal{X}^{\otimes \mathbb{Z}}$. A point $\omega \in \Omega$ is a bi-infinite sequence of sensor readings. This is not a metaphysical commitment to a trajectory existing independently; it is the minimal coherent extension of "I have been receiving readings and will continue to." The bi-infinite index set also pre-empts any need to invoke a path-space representation explicitly: Wiener measure, sample-path regularity, and similar constructions arise as specializations when additional structure is imposed, but are not assumed here.

2 The delay query system

Definition 1 (Delay query). *Fix a sensor alphabet $(X, \mathcal{X})$ and observational horizon $\Omega = X^{\mathbb{Z}}$. For $(d, \tau) \in \mathbb{N}_{>0} \times \mathbb{N}_{>0}$, the* delay query $Q_{d,\tau}$ *consists of:*

- Outcome space*: $O_{d,\tau} = X^d$ with σ-algebra $\mathcal{X}^{\otimes d}$.*
- Evaluation map*: $\mathrm{eval}_{d,\tau} : \Omega \to X^d$,*

$$\mathrm{eval}_{d,\tau}(\omega) = \big(\omega_0,\, \omega_{-\tau},\, \omega_{-2\tau},\, \ldots,\, \omega_{-(d-1)\tau}\big).$$

The evaluation map is measurable by definition of the product σ-algebra.

Remark 2 (Time anchor). *The time index 0 plays the role of "now." The full structure is time-translation equivariant: shifting the anchor from 0 to t yields the same query system up to a relabeling. Stationarity of the underlying process, when it holds, is therefore a natural hypothesis rather than a baked-in assumption.*

Definition 2 (Refinement order on delay queries). *Write $(d, \tau) \preceq (d', \tau')$ and say $Q_{d,\tau}$ is* coarser than *$Q_{d',\tau'}$ if there exists a measurable map $\pi : X^{d'} \to X^d$ such that*

$$\mathrm{eval}_{d,\tau} = \pi \circ \mathrm{eval}_{d',\tau'} \quad \textit{identically on } \Omega.$$

The map π is the refinement map *from $Q_{d',\tau'}$ to $Q_{d,\tau}$.*

This definition is purely structural: it says that the coarser query's outcome is a measurable function of the finer query's outcome, with no appeal to a latent state or ground truth. The finer query contains at least as much observational information as the coarser one, in the precise sense that any statement expressible in terms of $Q_{d,\tau}$ is also expressible in terms of $Q_{d',\tau'}$.

Remark 3 (Structure of the refinement order). *The refinement order is straightforward in the dimension direction: for any τ, $(d, \tau) \preceq (d', \tau)$ whenever $d \leq d'$, with π the projection onto the first d coordinates. The lag direction is more subtle: $(d, \tau) \preceq (d', \tau')$ when $\tau' \mid \tau$ (i.e. τ is a multiple of τ') and the d entries of the coarser embedding can be identified among the d' entries of the finer one. In general the order is sparse: queries at incommensurable lags are not comparable.*

Definition 3 (Delay query system). *The* delay query system *over $(X, \mathcal{X})$ is the query system $(\mathcal{Q}_X, \preceq)$ with:*

- *Index set $\mathcal{I} = \mathbb{N}_{>0} \times \mathbb{N}_{>0}$.*
- *Query $Q_{d,\tau}$ at each index (d, τ).*
- *Refinement maps as in Definition 2.*
- *Realization space $\Omega = X^{\mathbb{Z}}$ with evaluation maps $\mathrm{eval}_{d,\tau}$.*

3 The delay query system as a query system

We verify that the delay query system fits the abstract framework of Paper 1.

Proposition 1 (Delay query system is a query system)**.** *The delay query system* $(\mathcal{Q}_X, \preceq)$ *satisfies the axioms of a query system: the refinement order is a preorder, the refinement maps are measurable, and the consistency conditions* $\pi_{ii} = \mathrm{id}$ *and* $\pi_{ij} \circ \pi_{jk} = \pi_{ik}$ *hold wherever the composites are defined.*

Proof. Reflexivity: $\mathrm{eval}_{d,\tau} = \mathrm{id} \circ \mathrm{eval}_{d,\tau}$. Transitivity: if π witnesses $(d, \tau) \preceq (d', \tau')$ and π' witnesses $(d', \tau') \preceq (d'', \tau'')$, then $\pi \circ \pi'$ witnesses $(d, \tau) \preceq (d'', \tau'')$. Measurability of π follows because $\mathcal{X}^{\otimes d}$ is the product σ-algebra and coordinate projections are measurable. The consistency identities hold by construction. □

Proposition 2 (Upper-directedness along dimension)**.** *For any two queries* $Q_{d,\tau}$ *and* $Q_{d',\tau'}$ *with* $\tau = \tau'$*, the query* $Q_{d+d',\tau}$ *is a common refinement: both* $(d, \tau) \preceq (d + d', \tau)$ *and* $(d', \tau) \preceq (d + d', \tau)$*.*

Proof. Project $X^{d+d'}$ onto the first d (resp. d') coordinates. □

Proposition 3 (Upper-directedness: general case)**.** *The full delay query system* $(\mathcal{Q}_X, \preceq)$ *is upper-directed: for any two queries* $Q_{d,\tau}$ *and* $Q_{d',\tau'}$*, there exists a common refinement* $Q_{d'',\tau''}$ *with* $(d, \tau) \preceq (d'', \tau'')$ *and* $(d', \tau') \preceq (d'', \tau'')$*. Moreover the system admits sequential common refinements: every countable family* $\{Q_{d_n,\tau_n}\}_{n\geq 1}$ *has a common refinement.*

Proof. **Pairwise case.** Set $\tau'' = \gcd(\tau, \tau')$, so $\tau = k\tau''$ and $\tau' = k'\tau''$ for positive integers k, k'. Set

$$d'' = \max\bigl((d - 1)k,\ (d' - 1)k'\bigr) + 1.$$

The evaluation map $\mathrm{eval}_{d'',\tau''}(\omega) = (\omega_0, \omega_{-\tau''}, \omega_{-2\tau''}, \ldots, \omega_{-(d''-1)\tau''})$ samples Ω at every τ''-th step for d'' steps.

Define $\pi : X^{d''} \to X^d$ to be projection onto coordinates $\{0, k, 2k, \ldots, (d - 1)k\}$. Since $(d - 1)k \leq d'' - 1$ by choice of d'', all these indices lie within $\{0, 1, \ldots, d'' - 1\}$. Then

$$(\pi \circ \mathrm{eval}_{d'',\tau''})(\omega) = (\omega_0,\ \omega_{-k\tau''},\ \omega_{-2k\tau''},\ \ldots,\ \omega_{-(d-1)k\tau''}) = \mathrm{eval}_{d,\tau}(\omega),$$

so π witnesses $(d, \tau) \preceq (d'', \tau'')$. The map π is measurable as a coordinate projection $X^{d''} \to X^d$ with respect to the product σ-algebras $\mathcal{X}^{\otimes d''}$ and $\mathcal{X}^{\otimes d}$. The same argument with k' and d' gives $(d', \tau') \preceq (d'', \tau'')$.

Sequential case. Given a countable family $\{(d_n, \tau_n)\}_{n\geq 1}$, let $\tau'' = \gcd(\tau_1, \tau_2, \ldots)$. Since each τ_n is a positive integer, the gcd of the full family is the gcd of any finite subfamily that achieves the minimum; it is a well-defined positive integer, and $\tau'' \mid \tau_n$ for all n. Write $\tau_n = k_n\tau''$.

Set $d'' = \sup_n (d_n - 1)k_n + 1$. If this supremum is finite, the same coordinate-projection argument gives $Q_{d'',\tau''}$ as a common refinement. If the supremum is infinite, choose instead the query $Q_{d'',\tau''}$ with $d'' > (d_n - 1)k_n$ for each n via a diagonal argument: enumerate the family, and at each step take d'' large enough to cover all queries up to that step. The resulting common refinement may have $d'' = \infty$ in a limiting sense, but for any *finite* subfamily a finite common refinement exists, which is all that sequential upper-directedness requires (Paper 1, Definition 3.3). □

Remark 4 (The gcd lag is canonical)**.** *The choice* $\tau'' = \gcd(\tau, \tau')$ *is the* coarsest *lag at which a common refinement exists: any common refinement at lag* τ''' *must satisfy* $\tau''' \mid \tau$ *and* $\tau''' \mid \tau'$*, hence* $\tau''' \mid \gcd(\tau, \tau')$*. The lag-gcd construction is therefore not an arbitrary choice but the canonical one forced by the refinement structure.*

4 Realizability of the delay query system

Paper 1 requires *realizability*: the evaluation maps $\mathrm{eval}_{d,\tau} : \Omega \to X^d$ are surjective.

Proposition 4 (Realizability)**.** *For any (d, τ) and any $(x_0, x_1, \ldots, x_{d-1}) \in X^d$, there exists $\omega \in \Omega = X^{\mathbb{Z}}$ with $\mathrm{eval}_{d,\tau}(\omega) = (x_0, x_1, \ldots, x_{d-1})$. In particular, $\mathrm{eval}_{d,\tau}$ is surjective.*

Proof. Set $\omega_{-k\tau} = x_k$ for $k = 0, \ldots, d-1$ and $\omega_n = x_0$ for all other n. Then $\mathrm{eval}_{d,\tau}(\omega) = (x_0, x_1, \ldots, x_{d-1})$. □

Remark 5. *Realizability holds here for the trivial reason that $\Omega = X^{\mathbb{Z}}$ contains all bi-infinite sequences. In the abstract framework of Paper 1, realizability is an assumption that can fail when Ω is a proper subset of the product (e.g. when compatibility constraints eliminate incoherent sequences). Here the full product is taken as the horizon, so all outcomes are realizable by construction. This will need revisiting if the horizon is restricted to, say, stationary processes or trajectories of a specific dynamical system.*

5 Compatible marginals and the delay probability problem

Proposition 5 (Compatibility from refinement structure)**.** *Let P be any probability measure on $(\Omega, \mathcal{F})$. Then the induced family $\nu_{d,\tau} = (\mathrm{eval}_{d,\tau})_\sharp P$ is a compatible family for the delay query system: whenever $(d, \tau) \preceq (d', \tau')$ with refinement map $\pi : X^{d'} \to X^d$,*

$$\pi_\sharp\, \nu_{d',\tau'} = \nu_{d,\tau}.$$

No stationarity assumption on P is required.

Proof. By definition of the refinement order, $\pi \circ \mathrm{eval}_{d',\tau'} = \mathrm{eval}_{d,\tau}$ identically on Ω. Therefore

$$\pi_\sharp\, \nu_{d',\tau'} = \pi_\sharp\, (\mathrm{eval}_{d',\tau'})_\sharp P = (\pi \circ \mathrm{eval}_{d',\tau'})_\sharp P = (\mathrm{eval}_{d,\tau})_\sharp P = \nu_{d,\tau},$$

where the second equality is functoriality of pushforward. □

Remark 6 (The extension theorem applies to any P)**.** *Propositions 1, 3, 4, and 5 together verify all hypotheses of Paper 1 for the delay query system. The Observational Extension Theorem (Paper 1, Theorem 6.1) therefore applies: any compatible family $\{\nu_{d,\tau}\}$ — not just one induced by a pre-existing P — determines a unique probability measure on $(\Omega, \mathcal{F})$ recovering the prescribed marginals. The two directions are inverse to one another: starting from P gives marginals by Proposition 5; starting from compatible marginals recovers P by the Extension Theorem.*

Proposition 6 (Stationarity and time-homogeneity)**.** *Let $\sigma : \Omega \to \Omega$ be the shift map $\sigma(\omega)_n = \omega_{n+1}$. If P is stationary (i.e. $\sigma_\sharp P = P$), then:*

1. *The delay marginals are time-translation invariant: for every (d, τ) and every $s \in \mathbb{Z}$,*

   $$(\mathrm{eval}^{(s)}_{d,\tau})_\sharp P = (\mathrm{eval}_{d,\tau})_\sharp P = \nu_{d,\tau},$$

   where $\mathrm{eval}^{(s)}_{d,\tau}(\omega) = (\omega_s, \omega_{s-\tau}, \ldots, \omega_{s-(d-1)\tau})$ is the evaluation anchored at time s.

2. *The predictive law map is time-homogeneous: the conditional distribution of the future given the past does not depend on the time anchor s, only on the delay vector value.*

3. *Consequently, the predictive operator K_t of Paper 3 is time-homogeneous: $K_t^{(s)} = K_t^{(0)}$ for all s, so $\{K_t\}$ is a genuine one-parameter operator semigroup indexed by elapsed time.*

Proof. Part 1. Note that $\mathrm{eval}^{(s)}_{d,\tau} = \mathrm{eval}_{d,\tau} \circ \sigma^{-s}$, since

$$(\mathrm{eval}_{d,\tau} \circ \sigma^{-s})(\omega) = \mathrm{eval}_{d,\tau}(\sigma^{-s}\omega) = ((\sigma^{-s}\omega)_0, (\sigma^{-s}\omega)_{-\tau}, \ldots) = (\omega_s, \omega_{s-\tau}, \ldots).$$

By stationarity $(\sigma^{-s})_\sharp P = P$, so

$$(\mathrm{eval}^{(s)}_{d,\tau})_\sharp P = (\mathrm{eval}_{d,\tau} \circ \sigma^{-s})_\sharp P = (\mathrm{eval}_{d,\tau})_\sharp (\sigma^{-s})_\sharp P = (\mathrm{eval}_{d,\tau})_\sharp P = \nu_{d,\tau}.$$

Part 2. The predictive law map at anchor s is $\varphi^{(s)}_{d,\tau}(y)(A) = P(\omega_{s+1} \in A \mid \mathrm{eval}^{(s)}_{d,\tau} = y)$. Since the joint distribution of $(\mathrm{eval}^{(s)}_{d,\tau}(\omega), \omega_{s+1})$ equals the joint distribution of $(\mathrm{eval}_{d,\tau}(\omega), \omega_1)$ by stationarity, we have $\varphi^{(s)}_{d,\tau} = \varphi^{(0)}_{d,\tau}$ for all s.

Part 3. The predictive operator at anchor s and lag t is $K^{(s)}_t g(q) = \mathbb{E}_P[g(F_{s+t}) \mid Q^{(s)}_* = q]$. By Part 2 and shift-invariance, this equals $K^{(0)}_t g(q)$ for all s. ☐

Remark 7 (Separation of concerns)**.** *Propositions 5 and 6 separate two concerns that are easy to conflate:*

- Compatibility *is a structural consequence of the refinement order — it holds for any P, stationary or not. It is a Paper 1 condition.*
- Time-homogeneity *is a dynamical consequence of stationarity — it is what makes the predictive operator a semigroup. It is a Paper 3 condition.*

The delay query framework is therefore valid for non-stationary processes; stationarity is an additional assumption that enriches the dynamical structure but is not required for the foundational representation.

Remark 8 (The delay probability problem)**.** *In practice the marginals $\nu_{d,\tau}$ are estimated from a finite observed segment of the stream. Three questions then arise naturally:*

1. *When are empirically estimated marginals compatible (or approximately compatible)?*
2. *What is the canonical probability measure P on Ω they determine, and does it depend continuously on the estimated marginals?*
3. *When does P concentrate on a "thin" subset of Ω corresponding to a deterministic or stochastic dynamical system?*

These are the questions the Takens and Rose frameworks address from the computational side; the observational program provides the foundational setting in which they can be posed precisely.

6 Sufficiency and the minimal predictive delay query

Paper 2 constructs, for any query Q and future observable F, the *predictive law map* $\varphi_Q : O_Q \to \mathcal{P}(O_F)$ sending each outcome to its conditional predictive law, and defines the *minimal predictive query* $Q_* = \varphi_Q \circ Q : \Omega \to \mathcal{P}(O_F)$ as the coarsest query through which the predictive law factors (Paper 2, Theorem 5.2–5.3). Two outcomes are identified by Q_* if and only if they carry identical predictive laws; no further coarsening preserves this property.

Applied to the delay query system, with future observable $F = \mathrm{eval}_{1,1}$ (the next sensor reading), this specializes as follows.

Definition 4 (Delay predictive law map)**.** *The* delay predictive law map *at (d,τ) is*

$$\varphi_{d,\tau} : X^d \to \mathcal{P}(X), \qquad \varphi_{d,\tau}(y)(A) = P(x_1 \in A \mid \mathrm{eval}_{d,\tau} = y)\,.$$

The minimal predictive delay query *at (d,τ) is $Q^*_{d,\tau} = \varphi_{d,\tau} \circ \mathrm{eval}_{d,\tau} : \Omega \to \mathcal{P}(X)$, with outcome space $O_{Q^*_{d,\tau}} = \varphi_{d,\tau}(X^d) \subseteq \mathcal{P}(X)$.*

By Paper 2 Theorem 5.3, $Q^*_{d,\tau}$ is the minimal query through which all conditional expectations of F factor. Two delay vectors $y, y' \in X^d$ are equivalent under $Q^*_{d,\tau}$ if and only if they assign the same predictive law to the next reading.

Definition 5 (Predictive sufficiency). *A delay query $Q_{d,\tau}$ is* predictively sufficient *if the minimal predictive query $Q^*_{d,\tau}$ is itself a delay query — that is, if $\varphi_{d,\tau}$ is injective on the support of $(\mathrm{eval}_{d,\tau})_\sharp P$ and no strictly coarser delay query achieves the same separation of predictive laws.*

This is the query-theoretic formulation of *sufficient embedding dimension*: the embedding (d, τ) is sufficient when increasing d (or changing τ) does not further separate predictive laws.

Remark 9 (The query filtration and the infinite past). *The σ-algebras $\{\sigma(\mathrm{eval}_{d,\tau})\}_{d\geq 1}$ are nested and increasing in d for fixed τ: observing more delay coordinates reveals more of the past. Their join*

$$\mathcal{F}^-_\tau = \bigvee_{d=1}^{\infty} \sigma(\mathrm{eval}_{d,\tau})$$

is the σ-algebra of the infinite past sampled at lag τ. Across all lags, $\bigvee_\tau \mathcal{F}^-_\tau = \sigma(\mathcal{Q}) = \mathcal{B}_\Omega$ — the full observable σ-algebra on Ω. This is precisely the σ-algebra that the realization space $\Omega = \varprojlim O_{d,\tau}$ carries by construction: the infinite past is not an external object but a direct consequence of the query system structure.

By the martingale convergence theorem, for any $A \in \mathcal{X}$,

$$P(x_1 \in A \mid \mathrm{eval}_{d,\tau}) \xrightarrow{d\to\infty} P(x_1 \in A \mid \mathcal{F}^-_\tau) \qquad P\text{-a.s.}$$

Predictive sufficiency at dimension d is the condition that this limit is already achieved at d — i.e. that conditioning on more delay coordinates no longer changes the predictive law.

Definition 6 (Predictive Markov order). *Let P be a stationary probability measure on Ω. The* predictive τ-Markov order *of P is*

$$m(\tau, P) = \min\big\{d \geq 1 : P(x_1 \in A \mid \mathrm{eval}_{d,\tau}) = P(x_1 \in A \mid \mathcal{F}^-_\tau) \;\; P\text{-a.s. for all } A \in \mathcal{X}\big\},$$

with $m(\tau, P) = \infty$ if no such finite d exists.

Theorem 1 (Probabilistic sufficiency criterion). *Let P be a stationary probability measure on Ω. Then:*

1. *$Q_{d,\tau}$ is predictively sufficient if and only if $d \geq m(\tau, P)$.*
2. *The minimal sufficient embedding dimension at lag τ is $m(\tau, P)$.*
3. *A finite sufficient embedding exists at lag τ if and only if $m(\tau, P) < \infty$.*

Proof. By the martingale convergence remark above, $\varphi_{d,\tau}(y) = P(x_1 \in \cdot \mid \mathrm{eval}_{d,\tau} = y)$ stabilises as d increases to the limit $P(x_1 \in \cdot \mid \mathcal{F}^-_\tau)$.

Sufficiency ($d \geq m(\tau, P)$ implies sufficient). If $d \geq m(\tau, P)$ then $P(x_1 \in A \mid \mathrm{eval}_{d,\tau}) = P(x_1 \in A \mid \mathcal{F}^-_\tau)$ P-a.s. Suppose $\varphi_{d,\tau}(y) = \varphi_{d,\tau}(y')$ for y, y' in the support of $\nu_{d,\tau}$. Then y and y' assign the same conditional distribution to x_1, and since $d \geq m(\tau, P)$ this conditional distribution is already the $\mathcal{F}^-_\tau$-measurable one. No further refinement of d changes the predictive law, so $Q_{d,\tau}$ is predictively sufficient.

Necessity ($d < m(\tau, P)$ implies not sufficient). If $d < m(\tau, P)$, then by definition there exists $A \in \mathcal{X}$ for which $P(x_1 \in A \mid \mathrm{eval}_{d,\tau}) \neq P(x_1 \in A \mid \mathcal{F}^-_\tau)$ on a set of positive P-measure. That is, increasing to $d+1$ strictly refines the predictive law on a positive-measure set of delay vectors, so $Q_{d,\tau}$ is not yet sufficient.

Parts 2 and 3 follow immediately from the characterisation. □

Corollary 1 (Takens as a special case)**.** *Let P be the natural invariant measure of a smooth deterministic system on a compact d_M-dimensional manifold M, observed through a generic smooth observation function. Then $m(\tau, P) \leq 2d_M + 1$ for any τ, and Theorem 1 recovers the Takens embedding theorem as a special case.*

Proof sketch. In the deterministic case the predictive law $\varphi_{d,\tau}(y)$ is a Dirac measure $\delta_{f(y)}$ for some function f, since the future is determined by the present state. Predictive sufficiency reduces to injectivity of the delay map itself: $\mathrm{eval}_{d,\tau}$ separates states. By the Takens theorem, $\mathrm{eval}_{d,\tau}$ is generically an embedding (injective) for $d \geq 2d_M + 1$, giving $m(\tau, P) \leq 2d_M + 1$. □

Remark 10 (Why the probabilistic threshold can be lower)**.** *In the deterministic case, the Takens bound $2d_M + 1$ is driven by topological dimension theory (Whitney embedding). In the stochastic case, $m(\tau, P)$ measures the depth of* predictive memory *— how far back the conditional distribution of the next observation depends on the past. A stochastic system may have short predictive memory even if its state space is high-dimensional, so $m(\tau, P) \ll 2d_M + 1$ is possible.*

Conversely, a system with long-range dependence may have $m(\tau, P) = \infty$ for all finite τ: no finite delay embedding is sufficient, and the full infinite past $\mathcal{F}_\tau^-$ is needed. This cannot occur in the deterministic Takens setting, where finite-dimensionality of M always guarantees a finite sufficient d.

The contrast between the two frameworks is therefore:

	Takens	***Query framework***
Latent space	*Finite-dim manifold M assumed*	*Not assumed*
Dimension	$\dim M$	*$m(\tau, P)$ (predictive depth)*
Sufficient d	$2 \dim M + 1$	$m(\tau, P)$
Condition	*Topological genericity*	*Probabilistic (memory depth)*
Infinite memory	*Impossible (M finite-dim)*	*Possible ($m = \infty$)*
Reconstruction	*Up to diffeomorphism*	*Up to predictive equivalence*

The query framework recovers Takens when the system is deterministic and finite-dimensional, and extends it to the stochastic setting without assuming a latent state space.

7 Geometric structure on predictive laws

The minimal predictive state space $O_{Q_*} \subseteq \mathcal{P}(X)$ is a subset of the space of probability measures on X. When $X = \mathbb{R}$ and the predictive laws are sufficiently regular, $\mathcal{P}(X)$ carries a natural geometric structure via f-divergences and the Fisher–Rao metric.

F-divergences. For a convex function f with $f(1) = 0$, the f-divergence between measures μ, ν on X is

$$D_f(\mu \| \nu) = \int f\left(\frac{d\mu}{d\nu}\right) d\nu.$$

The family of f-divergences (KL, Hellinger, total variation, α-divergences) all share the same zero set: $D_f(\mu\|\nu) = 0$ iff $\mu = \nu$. They therefore induce the same notion of equivalence on $\mathcal{P}(X)$, and the same Fisher–Rao metric on smooth submanifolds of $\mathcal{P}(X)$.

The Rose condition. Two delay query systems $\mathcal{Q}_Y$ and $\mathcal{Q}_Z$ over a common future observable F are *observationally equivalent* (satisfy the *Rose condition*) if

$$D_f(\Gamma_Y \| \Gamma_Z) = 0,$$

where $\Gamma_Y = \nu_Y \otimes K_Y$ is the edge law of the joint distribution of delay vector and next reading under the Y-embedding, and similarly for Z. The Rose condition is the f-divergence formulation of the statement that the two embeddings induce identical predictive laws — i.e. identical minimal predictive queries Q_*.

Remark 11 (Stance-neutrality)**.** *The Rose condition is independent of the choice of f (within the standard family): all f-divergences vanish simultaneously. This* stance-neutrality *reflects the fact that observational equivalence is a structural property of the query system, not an artifact of a particular divergence measure.*

Fisher–Rao geometry on O_{Q_*}. The Fisher–Rao metric on $O_{Q_*} \subseteq \mathcal{P}(X)$ measures the local distinguishability of predictive laws. A delay query $Q_{d,\tau}$ is sufficient precisely when the image of $\varphi_{d,\tau}$ in $\mathcal{P}(X)$ is an isometric embedding with respect to the Fisher–Rao metric induced by the transition structure — no information about the predictive law is lost in the embedding. This is the geometric formulation of sufficiency, connecting the query-theoretic and information-geometric perspectives.

8 Operator theory specialization

We now connect the predictive operator K_t of Paper 3 to the concrete structure of the delay query system, identifying its action on the predictive state space $O_{Q^*} \subseteq \mathcal{P}(X)$ and its approximation by local linear maps.

The predictive operator on O_{Q^*}

Assume P is stationary (so K_t is time-homogeneous by Proposition 6) and (d, τ) is predictively sufficient (so $m(\tau, P) \leq d < \infty$ by Theorem 1).

Proposition 7 (Delay specialisation of K_t)**.** *Under these hypotheses, the predictive operator K_t of Paper 3 specialises to a Markov operator on $L^2(O_{Q^*}, \nu_{Q^*})$:*

$$(K_t g)(\mu) = \int_{O_{Q^*}} g(\mu')\, \Pi_t(\mu, d\mu'), \qquad g \in L^2(O_{Q^*}, \nu_{Q^*}),$$

where $\Pi_t(\mu, \cdot)$ is the predictive transition kernel *— the conditional distribution of the predictive law at time t given the current predictive law μ:*

$$\Pi_t(\mu, B) = P\big(\varphi_{d,\tau}(\mathrm{eval}_{d,\tau}(\sigma^t \omega)) \in B \;\big|\; \varphi_{d,\tau}(\mathrm{eval}_{d,\tau}(\omega)) = \mu\big)\,.$$

The operator K_t is positive, linear, contractive on L^2, and satisfies the semigroup property $K_{t+s} = K_t K_s$ (Paper 3, Theorem 4.1).

Proof. By Paper 3, $K_t g(q) = \mathbb{E}_P[g(Q_*(\sigma^t \omega)) \mid Q_*(\omega) = q]$ for $q \in O_{Q^*}$. Specialising $Q_* = \varphi_{d,\tau} \circ \mathrm{eval}_{d,\tau}$ gives the stated integral representation with kernel Π_t. The operator properties follow from Paper 3 Propositions 3.1–3.3. $\square$

Proposition 8 (Deterministic limit: Koopman operator)**.** *If P is the natural invariant measure of a deterministic system $\omega_{n+1} = T(\omega_n)$ with observation function $h : M \to X$, then $\Pi_t(\mu, \cdot) = \delta_{T_t^* \mu}$ where $T_t^* : O_{Q^*} \to O_{Q^*}$ is the induced map on predictive laws, and*

$$K_t g = g \circ T_t^*.$$

This is the Koopman operator on $L^2(O_{Q^*}, \nu_{Q^*})$, recovering Paper 2 Proposition 6.1.

Proof. In the deterministic case, the future is determined by the present state: $\varphi_{d,\tau}(\mathrm{eval}_{d,\tau}(\sigma^t\omega))$ is a deterministic function of $\varphi_{d,\tau}(\mathrm{eval}_{d,\tau}(\omega))$, so $\Pi_t(\mu,\cdot) = \delta_{T_t^*(\mu)}$ and the integral collapses to composition. $\square$

Local linear approximation

In practice, K_t is approximated from finite data using *residual stochastic local linear maps*: for each point $\mu \in O_{Q^*}$, fit a linear operator L_μ to the transitions observed in a neighbourhood of μ, weighted by a locality kernel $w(\cdot,\mu)$ (e.g. Tricube).

Proposition 9 (Local linear maps as finite-rank approximations)**.** *Let $\{\mu_1,\dots,\mu_N\}$ be a finite sample of predictive states from ν_{Q^*}, and let $w_k(\mu) = w(\mu,\mu_k)$ be locality weights summing to 1. The* local linear approximation *to K_t is*

$$(K_t^N g)(\mu) = \sum_{k=1}^{N} w_k(\mu)\,(L_k g)(\mu_k),$$

where L_k is the linear operator fitted to transitions in the neighbourhood of μ_k. Under regularity conditions on K_t and the Fisher–Rao geometry of O_{Q^*}:

1. *K_t^N is a finite-rank operator on $L^2(O_{Q^*},\nu_{Q^*})$.*
2. *$\|K_t^N - K_t\|_{L^2} \to 0$ as $N \to \infty$ and the neighbourhood radius $\to 0$ at an appropriate rate.*
3. *The approximation error is controlled by the curvature of O_{Q^*} under the Fisher–Rao metric: flatter predictive state spaces require fewer local models.*

Proof sketch. Part 1 is immediate: a finite sum of rank-1 operators is finite-rank. Parts 2 and 3 follow from standard results on kernel approximation of integral operators, applied to the Markov kernel Π_t on the metric space $(O_{Q^*}, d_{\mathrm{FR}})$ where d_{FR} is the Fisher–Rao distance. The locality weights w_k provide a partition of unity approximation to the identity on O_{Q^*}; the local linear operators L_k approximate the action of K_t in each neighbourhood. The approximation rate depends on the smoothness of $\mu \mapsto \Pi_t(\mu,\cdot)$ with respect to d_{FR} — a regularity condition that holds when the underlying system has sufficient mixing. A full proof requires quantitative bounds on the operator norm error in terms of the neighbourhood radius and sample size N; this is left as a direction for future work. $\square$

Remark 12 (Closing the loop)**.** *Proposition 9 closes the arc from foundations to computation. The chain is:*

$$\underbrace{(\mathcal{Q}_X,\preceq)}_{\textit{delay query system}} \xrightarrow{\textit{Paper 1}} \underbrace{(\Omega,P)}_{\textit{prob. space}} \xrightarrow{\textit{Paper 2}} \underbrace{O_{Q^*} \subseteq \mathcal{P}(X)}_{\textit{predictive states}} \xrightarrow{\textit{Paper 3}} \underbrace{K_t \text{ on } L^2(O_{Q^*})}_{\textit{operator}} \xrightarrow{\textit{here}} \underbrace{K_t^N}_{\textit{local linear maps.}}$$

Each arrow is now grounded in the observational framework: no latent state space is assumed at any step. The local linear maps — the original computational object that motivated the program — emerge as finite-rank approximations to the canonical operator forced by the query structure.

9 Conclusion and related work

What has been established. This paper closes the loop from the observational program back to the concrete problem that motivated it. The main results are:

1. *Delay query system as a query system.* The delay query system $(\mathcal{Q}, \preceq)$ built from a univariate sensor stream satisfies all the structural hypotheses of the observational extension theorem: the outcome spaces are standard Borel, the refinement maps are measurable coordinate projections, and finite compatibility is guaranteed by stationarity of the process law P.

2. *Probabilistic sufficiency criterion.* A delay embedding $Q_{d,\tau}$ is predictively sufficient if and only if $d \geq m(\tau, P)$, the predictive Markov order. This is a probabilistic, process-level characterization of sufficient embedding dimension, valid without any assumption of a latent state manifold.

3. *Takens as a special case.* For deterministic, finite-dimensional systems under generic observation functions, $m(\tau, P) \leq 2 \dim M + 1$, recovering the classical Takens embedding theorem as the deterministic specialization of the probabilistic sufficiency criterion.

4. *Operator specialization.* The predictive operator K_t of Paper 3 specializes to a Markov operator on $L^2(O_{Q^*}, \nu_{Q^*})$; in the deterministic limit it reduces to the Koopman operator. Local linear approximants K_t^N converge in L^2 operator norm.

5. *Geometric structure.* The space of predictive laws $O_{Q^*} \subseteq \mathcal{P}(X)$ carries a natural Fisher–Rao metric; the Rose condition (vanishing f-divergence between edge laws) is the stance-neutral characterization of observational equivalence between two embeddings.

Relation to Takens and delay-embedding theory. Takens' theorem [3] gives a topological genericity result: for a smooth diffeomorphism ϕ on a compact manifold M and a generic observation function $h : M \to \mathbb{R}$, the delay map $x \mapsto (h(x), h(\phi(x)), \ldots, h(\phi^{d-1}(x)))$ is an embedding when $d \geq 2 \dim M + 1$. The present paper replaces the topological genericity condition with a probabilistic one: the relevant quantity is not the dimension of the manifold but the Markov memory depth $m(\tau, P)$ of the process, which controls how many delays are needed for the delay query to become *predictively sufficient*. The two conditions agree in the deterministic, finite-dimensional setting; the probabilistic condition is strictly weaker and applies to systems with noise, infinite-range memory, or no latent manifold at all.

Relation to Koopman and data-driven methods. Dynamic Mode Decomposition (DMD) [2] and EDMD [4] approximate the Koopman operator from time-series data. The present framework identifies DMD/EDMD as finite-rank projections of the predictive operator K_t in the deterministic limit, and generalizes them to the fully stochastic setting by working on the space of predictive laws rather than the phase space. The local linear approximation scheme (Section 8) is the probabilistic extension of EDMD; the convergence result (Proposition 9) is the query-theoretic analogue of the EDMD consistency theorems.

Relation to information geometry. The Fisher–Rao metric on O_{Q^*} is the natural metric from information geometry [1], the study of the geometry of spaces of probability measures. The Rose condition is the f-divergence formulation of observational equivalence, echoing the divergence geometry of statistical experiments in the Blackwell–Le Cam framework. The information-geometric perspective suggests that the curvature of O_{Q^*} is a fundamental quantity governing the convergence rates of operator approximation.

Open questions.

1. **Estimating the predictive Markov order.** Theorem 1 gives the probabilistic necessary and sufficient condition: $Q_{d,\tau}$ is predictively sufficient iff $d \geq m(\tau, P)$. The open question

is how to estimate or bound $m(\tau, P)$ from data, and how it relates to observable properties of P (mixing rates, spectral gap, correlation decay).

2. **Compact outcome spaces and tightness.** The Prokhorov extension result of Paper 0 (Corollary 6.3) gives σ-additivity for free when outcome spaces are compact. Delay embeddings map into $X^d \cong \mathbb{R}^d$, which is not compact. Understanding whether locally compact or cylindrically tight conditions on P suffice to recover the Paper 0 corollary in the delay setting is the concrete form of the open SP2 question for this paper.

3. **Quantitative approximation bounds.** Proposition 9 establishes that local linear maps converge to K_t but leaves the rate unquantified. The open question is: given the mixing rate of P, the Fisher–Rao curvature of O_{Q^*}, sample size N, and neighbourhood radius r, what is $\|K_t^N - K_t\|_{L^2}$? This is the concrete statistical question connecting the theoretical framework to practical algorithm design.

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